The existing TF-IDF cosine-similarity option does work well to recommend songs of the same genre using their lyrics as data. As we saw throughout the lab, the inner product and covariance methods were flawed as their ranking system was heavily influenced by the sizes of the song vectors. Songs with high word counts generally ranked high with these methods, as the methods do not rescale/normalize vectors. However, both the correlation and cosine-similarity processes normalize the results. This was reflected in the genre precision testing, as the raw inner product correctly provided song recommendations of the same genre well below the accuracy of the regular cosine similarity method (9.3% and 19.1% respectively). Adding in the TF-IDF weights also seemed to make a difference, as the TF-IDF cosine-similarity precision test (21.1% success rate at picking songs with the same genre) did better than just the regular cosine-similarity test. This makes sense, as it lowers the effect of common word matches and emphasizes rarer words that are more indicative of similarity. For example, a song that has many “and”, “the”, or “I” matches is less indicative of song similarity since so many songs have these words.

Due to the similarity of correlation and cosine-similarity, it was reasonable to think that correlation could perform better. Both normalize and both generated very similar recommendations in the lab. Further, the correlation method centers the word values by subtracting the mean from the word values before normalizing the result, which may result in higher accuracy. Due to the success of the TF-IDF in the cosine-similarity precision test, it was added to the correlation method as well. After running the TF-IDF correlation precision test (success rate of 19.8%), it was determined that the TF-IDF cosine similarity was still the superior method.

That does not mean that the TF-IDF cosine-similarity method cannot be improved or out-performed. Other suggestions to improve its success rate (using only lyrics as a data input) include using a hybrid TF-IDF/genre-specific word weight system with cosine-similarity. For example, higher weights could be applied to words more indicative of genres (“beer” and “truck” could have higher weights due to their association with country, “lonesome” or “blue” for the blues, “rock” and “roll” for rock & roll etc.). Using part of the TF-IDF weight system, lower weights would still be applied to very common basic words. However, the difference is that the common genre-specific words would have the highest weights (even more than the rare words: a rock song and a jazz song could both still have the word “zebra” but songs with “death” in them are more likely to be metal). Using this new weighted system along with the cosine-similarity, a more accurate song recommender could possibly be developed.